

Generation Interconnection System Impact Study Report

For Queue Project Q22-2239

Lovely 138 kV
September 15, 2024

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q22-2239. The Interconnection Customer proposes a 71.1 MW battery energy storage facility interconnecting at the Lovely 138 kV substation in the Kentucky area. The request was received on November 24, 2022. The study evaluated system intact and single contingency conditions using the regional planning case, including 502 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 1 overloaded facility. The total estimated cost of the network upgrades allocated to this project is \$143,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
DAVIS RD / SPALL NS 138 KV	HUBBARDSTOWN / POPLAR BLUFF GENERATING STA 345 KV	104.1	225	5.2

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
DAVIS RD / SPALL NS 138 KV	Reconductor line	\$2,474,000	6% (\$143,000)
	Total allocated to project		\$143,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.